UNSW SYDNEY

MATH2601 – Higher Linear Algebra

Topic 4 – Inner product spaces
Lecture 4.1 – Dot product and inner products

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Multilinear maps

Definition. Given vector spaces U, V, and W each over a field $\mathbb{F}$, a bilinear map is a function $T : U \times V \to W$ which is linear in each argument. That is, for all $\mathbf{u}, \mathbf{u}_i \in U$, $\mathbf{v}, \mathbf{v}_i \in V$, and $\lambda \in \mathbb{F}$, we have

$$T(\mathbf{u}_1 + \lambda \mathbf{u}_2,\ \mathbf{v}) = T(\mathbf{u}_1, \mathbf{v}) + \lambda T(\mathbf{u}_2, \mathbf{v}), \text{ and}$$
$$T(\mathbf{u}_1,\ \mathbf{v}_1 + \lambda \mathbf{v}_2) = T(\mathbf{u}, \mathbf{v}_1) + \lambda T(\mathbf{u}, \mathbf{v}_2).$$

We say T is bilinear on V if $T : V \times V \to W$ is bilinear.

This definition of bilinearity can be extended in general to multilinear maps.

Example. The real dot product $\mathbf{v} \cdot \mathbf{w}$ is a bilinear map $\mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$. The scalar triple product $(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{w}$ is a trilinear map $\mathbb{R}^n \times \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$.

Definition. A bilinear map T on V is symmetric if $T(\mathbf{u}, \mathbf{v}) = T(\mathbf{v}, \mathbf{u})$ for all $\mathbf{u}, \mathbf{v} \in V$. A multilinear map is symmetric if the output of T is unchanged regardless of the permutation of the arguments.

Definition. A bilinear map T on V is alternating if $T(\mathbf{u}, \mathbf{v}) = -T(\mathbf{v}, \mathbf{u})$ for all $\mathbf{u}, \mathbf{v} \in V$. A multilinear map is alternating if the output of T changes sign whenever two arguments swap position.

Example. The real dot product is symmetric, while the scalar triple product is alternating.

The real dot product

Definition. Recall that the real dot product is a (symmetric) bilinear map on $\mathbb{R}^n$ given by $\mathbf{u} \cdot \mathbf{v} = \mathbf{u}^T \mathbf{v} = u_1 v_1 + u_2 v_2 + \cdots + u_n v_n$.

Another characterising property of the real dot product is its connection to length of a geometric vector.

Definition. The length of a vector $\mathbf{v} \in \mathbb{R}^n$ is the scalar $||\mathbf{v}|| = \sqrt{\mathbf{v} \cdot \mathbf{v}}$.

A useful property of length is that a vector's length is 0 if and only if the vector is $\mathbf{0}$. We can in fact think of this as a property of the real dot product:

Definition. A bilinear map $T : \mathbb{F}^n \times \mathbb{F}^n \to \mathbb{F}$ is called positive definite if for all $\mathbf{v} \in \mathbb{F}^n$, we have $T(\mathbf{v}, \mathbf{v}) \geq 0$, with equality if and only if $\mathbf{v} = \mathbf{0}$.

Certainly the real dot product is positive definite (since it can be expressed as a sum of real squares).

Note that this notion of positive definiteness doesn't carry over if we try to extend the definition of the dot product to vectors in $\mathbb{C}^n$, nor to vectors in finite spaces $\mathbb{F}^n$, since in both cases there isn't a proper notion of inequality in the underlying scalar field (e.g. in $\mathbb{C}$, i is not comparable with 0, and in $\mathbb{Z}_2$, $1 = -1$ isn't comparable with 0). So for now we will just investigate properties of the real dot product.

Cauchy-Schwarz and triangle inequalities (real vectors)

Theorem. (Cauchy-Schwarz inequality for $\mathbb{R}^n$)
For all $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$, we have $-||\mathbf{u}||\,||\mathbf{v}|| \leq \mathbf{u} \cdot \mathbf{v} \leq ||\mathbf{u}||\,||\mathbf{v}||$.

Proof. Equality obviously holds when $\mathbf{u}$ or $\mathbf{v}$ is $\mathbf{0}$.

Otherwise, set $\mathbf{w} = \dfrac{\mathbf{u}}{||\mathbf{u}||} - \dfrac{\mathbf{v}}{||\mathbf{v}||}$. Then

$$\mathbf{w} \cdot \mathbf{w} = \frac{\mathbf{u} \cdot \mathbf{u}}{||\mathbf{u}||^2} - 2\frac{\mathbf{u} \cdot \mathbf{v}}{||\mathbf{u}||\,||\mathbf{v}||} + \frac{\mathbf{v} \cdot \mathbf{v}}{||\mathbf{v}||^2} = 2 - 2\frac{\mathbf{u} \cdot \mathbf{v}}{||\mathbf{u}||\,||\mathbf{v}||}.$$

Since $\mathbf{w} \cdot \mathbf{w} \geq 0$, we have $2 \geq 2\frac{\mathbf{u} \cdot \mathbf{v}}{||\mathbf{u}||\,||\mathbf{v}||}$ which rearranges to give $\mathbf{u} \cdot \mathbf{v} \leq ||\mathbf{u}||\,||\mathbf{v}||$ since $||\mathbf{u}||\,||\mathbf{v}|| > 0$.

An analogous argument for $\mathbf{w}' = \dfrac{\mathbf{u}}{||\mathbf{u}||} + \dfrac{\mathbf{v}}{||\mathbf{v}||}$ provides the other inequality.

Theorem. (Triangle inequality for $\mathbb{R}^n$)
For all $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$, we have $||\mathbf{u} + \mathbf{v}|| \leq ||\mathbf{u}|| + ||\mathbf{v}||$.

Proof. Note that $||\mathbf{u} + \mathbf{v}||^2 = (\mathbf{u} + \mathbf{v}) \cdot (\mathbf{u} + \mathbf{v}) = ||\mathbf{u}||^2 + 2\mathbf{u} \cdot \mathbf{v} + ||\mathbf{v}||^2$, so by Cauchy-Schwarz $||\mathbf{u} + \mathbf{v}||^2 \leq ||\mathbf{u}||^2 + 2||\mathbf{u}||\,||\mathbf{v}|| + ||\mathbf{v}||^2 = (||\mathbf{u}|| + ||\mathbf{v}||)^2$. Since both squared terms are positive, we can take square roots of both sides, yielding the triangle inequality.

Angles and orthogonality

Definition. The angle between two vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ is the number $\theta \in \mathbb{R}$ with $0 \leq \theta \leq \pi$ for which

$$\cos(\theta) = \frac{\mathbf{u} \cdot \mathbf{v}}{||\mathbf{u}|| \, ||\mathbf{v}||}.$$

Notice that by the Cauchy-Schwarz inequality, $-1 \leq \frac{\mathbf{u} \cdot \mathbf{v}}{||\mathbf{u}|| \, ||\mathbf{v}||} \leq 1$, so this notion of angle is well-defined.

Definition. Two vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ are orthogonal if and only if their dot product is 0 (that is, $\mathbf{u} \cdot \mathbf{v} = 0$).

Definition. An orthogonal set is any set $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\} \subseteq \mathbb{R}^n$ whose elements are pairwise orthogonal, that is $\mathbf{v}_i \cdot \mathbf{v}_j = 0$ for all $i \neq j$.

Note that $\mathbf{0}$ is orthogonal to all vectors according to this definition. Some texts sometimes exclude $\mathbf{0}$ from the definition of an orthogonal set. We will always specify if the set contains only nonzero vectors.

Definition. An orthogonal set $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\} \subseteq \mathbb{R}^n$ is orthonormal if every element has length 1, that is $\mathbf{v}_i \cdot \mathbf{v}_j = \begin{cases} 0 & \text{if } i \neq j \\ 1 & \text{if } i = j \end{cases}$ for all i, j.

Example. The standard basis in $\mathbb{R}^n$ is orthonormal.

Orthogonal complement (real vectors)

Lemma. Any orthogonal set $S \subseteq \mathbb{R}^n$ of nonzero vectors is linearly independent.

Proof. Let $S = \{\mathbf{v}_1, \dots, \mathbf{v}_k\}$. Suppose $\lambda_1 \mathbf{v}_1 + \cdots + \lambda_k \mathbf{v}_k = \mathbf{0}$ for some $\lambda_i \in \mathbb{R}$. Then for any particular i between 1 and k, taking the dot product with $\mathbf{v}_i$ on both sides of the equation gives

$$\begin{aligned}(\lambda_1 \mathbf{v}_1 + \cdots + \lambda_k \mathbf{v}_k) \cdot \mathbf{v}_i &= \mathbf{0} \cdot \mathbf{v}_i, \\ 0 + \cdots + 0 + \lambda_i \mathbf{v}_i \cdot \mathbf{v}_i + 0 + \cdots + 0 &= 0, \\ \lambda_i ||\mathbf{v}_i||^2 &= 0.\end{aligned}$$

Since $\mathbf{v}_i \neq \mathbf{0}$, we know $||\mathbf{v}_i|| \neq 0$ and so $\lambda_i = 0$ for all i. Hence S is linearly independent.

Definition. Given a subspace $X \leq \mathbb{R}^n$, its orthogonal complement in $\mathbb{R}^n$, denoted $X^\perp$, is the set $X^\perp = \{\mathbf{y} \in \mathbb{R}^n : \mathbf{y} \cdot \mathbf{x} = 0 \text{ for all } \mathbf{x} \in X\}$.

Lemma. For any subspace $X \leq \mathbb{R}^n$, we have $X^\perp \leq \mathbb{R}^n$.

Proof. Certainly $\mathbf{0} \in X^\perp$, so $X^\perp \neq \varnothing$.
Now for any $\mathbf{u}, \mathbf{v} \in X^\perp$ and $\lambda \in \mathbb{R}$, we have $\mathbf{u} \cdot \mathbf{x} = \mathbf{v} \cdot \mathbf{x} = 0$ for all $\mathbf{x} \in X$.
So $(\mathbf{u} + \lambda \mathbf{v}) \cdot \mathbf{x} = \mathbf{u} \cdot \mathbf{x} + \lambda \mathbf{v} \cdot \mathbf{x} = 0 + \lambda 0 = 0$, meaning $\mathbf{u} + \lambda \mathbf{v} \in X^\perp$.
Hence $X^\perp \leq \mathbb{R}^n$ by the subspace lemma.

The (standard) dot product

We next want to extend the definition of the dot product to work over $\mathbb{C}$ without violating positive definiteness.

Notation. For any $z \in \mathbb{C}$, the complex conjugate of z is denoted $\overline{z}$, found by changing the sign of the imaginary part (so $\overline{z} = z - 2\,\mathrm{Im}(z)$).

One convenient property of the complex conjugate is that $\overline{z}z = |z|^2 \geq 0$.

Definition. The (standard) dot product on $\mathbb{C}^n$ is a bilinear map given by

$$\mathbf{u} \cdot \mathbf{v} = \overline{\mathbf{u}}^T \mathbf{v} = \overline{u_1} v_1 + \overline{u_2} v_2 + \cdots + \overline{u_n} v_n.$$

Example. We have $\begin{pmatrix} 1+i \\ i \end{pmatrix} \cdot \begin{pmatrix} 1 \\ i \end{pmatrix} = (1-i)(1) + (-i)(i) = 2 - i.$

Lemma. The standard dot product is positive definite.

Proof. For any $\mathbf{v} \in \mathbb{C}^n$, we have $\mathbf{v} \cdot \mathbf{v} = |v_1|^2 + \cdots + |v_n|^2 \geq 0$, with $\mathbf{v} \cdot \mathbf{v} = 0$ if and only if $\mathbf{v} = \mathbf{0}$.

Definition. As before, the length of a vector $\mathbf{v} \in \mathbb{C}^n$ is $||\mathbf{v}|| = \sqrt{\mathbf{v} \cdot \mathbf{v}}$.

Notation. For any $\mathbf{v} \in \mathbb{C}^n$, the conjugate transpose (or Hermitian or adjoint) of $\mathbf{v}$ is denoted $\mathbf{v}^* = \overline{\mathbf{v}}^T$ (also sometimes written $\mathbf{v}^H$).

So we will often write $\mathbf{u} \cdot \mathbf{v} = \mathbf{u}^* \mathbf{v}$.

Properties of the standard dot product

Lemma. The standard dot product is conjugate symmetric, meaning for all $\mathbf{u}, \mathbf{v} \in \mathbb{C}^n$, we have $\mathbf{u} \cdot \mathbf{v} = \overline{\mathbf{v} \cdot \mathbf{u}}$.

Proof. Using the facts complex conjugation is additive, multiplicative, and self-inverse, we have

$$\begin{aligned} \overline{\mathbf{v} \cdot \mathbf{u}} &= \overline{\overline{v_1} u_1 + \overline{v_2} u_2 + \cdots + \overline{v_n} u_n} \\ &= v_1 \overline{u_1} + v_2 \overline{u_2} + \cdots + v_n \overline{u_n} \\ &= \mathbf{u} \cdot \mathbf{v}. \end{aligned}$$

Lemma. The standard dot product is linear in the second argument, meaning for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathbb{C}^n$ and $\lambda \in \mathbb{C}$, we have $\mathbf{u} \cdot (\mathbf{v} + \lambda \mathbf{w}) = \mathbf{u} \cdot \mathbf{v} + \lambda \mathbf{u} \cdot \mathbf{w}$.

Proof. We have $\mathbf{u} \cdot (\mathbf{v} + \lambda \mathbf{w}) = \mathbf{u}^*(\mathbf{v} + \lambda \mathbf{w}) = \mathbf{u}^* \mathbf{v} + \lambda \mathbf{u}^* \mathbf{w} = \mathbf{u} \cdot \mathbf{v} + \lambda \mathbf{u} \cdot \mathbf{w}$, using the linearity of matrix multiplication.

Lemma. The standard dot product is conjugate linear in the first argument, meaning for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathbb{C}^n$ and $\lambda \in \mathbb{C}$, we have $(\mathbf{v} + \lambda \mathbf{w}) \cdot \mathbf{u} = \mathbf{v} \cdot \mathbf{u} + \overline{\lambda} \mathbf{w} \cdot \mathbf{u}$.

Proof. Here $(\mathbf{v} + \lambda \mathbf{w}) \cdot \mathbf{u} = \overline{\mathbf{u} \cdot (\mathbf{v} + \lambda \mathbf{w})} = \overline{\mathbf{u} \cdot \mathbf{v} + \lambda \mathbf{u} \cdot \mathbf{w}} = \mathbf{v} \cdot \mathbf{u} + \overline{\lambda} \mathbf{w} \cdot \mathbf{u}$, using the facts complex conjugation is additive and multiplicative.

Because of the above two properties, the standard dot product is sometimes called sesquilinear.

Inner products

For the rest of Topic 4, the symbol $\mathbb{F}$ will represent only the fields $\mathbb{R}$ or $\mathbb{C}$.

Definition. An inner product on a vector space $(V, \mathbb{F})$ is a function $\langle\cdot\rangle : V \times V \to \mathbb{F}$ which satisfies the following properties for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$ and $\lambda \in \mathbb{C}$:

- Linear in the second argument: $\langle \mathbf{u}, \mathbf{v} + \lambda\mathbf{w}\rangle = \langle \mathbf{u}, \mathbf{v}\rangle + \lambda\langle \mathbf{u}, \mathbf{w}\rangle$.
- Conjugate symmetric: $\langle \mathbf{u}, \mathbf{v}\rangle = \overline{\langle \mathbf{v}, \mathbf{u}\rangle}$.
- Positive definite: $\langle \mathbf{v}, \mathbf{v}\rangle \in \mathbb{R}_0^+$ with $\langle \mathbf{v}, \mathbf{v}\rangle = 0$ if and only if $\mathbf{v} = \mathbf{0}$.

(Warning: Some texts instead define the inner product to be linear in the first argument! We will consistently use the above definition for this course.)

The vector space V equipped with $\langle\cdot\rangle$ is called an inner product space.

Notice that the first defining property of an inner product can also be broken into two simpler properties: preserves addition ($\langle \mathbf{u}, \mathbf{v} + \mathbf{w}\rangle = \langle \mathbf{u}, \mathbf{v}\rangle + \langle \mathbf{u}, \mathbf{w}\rangle$) and preserves scalar multiplication ($\langle \mathbf{u}, \lambda\mathbf{v}\rangle = \lambda\langle \mathbf{u}, \mathbf{v}\rangle$). (Proof: Exercise!)

Definition. In an inner product space V, the norm of a vector $\mathbf{v} \in V$ is denoted $||\mathbf{v}||$ and given by $||\mathbf{v}|| = \sqrt{\langle \mathbf{v}, \mathbf{v}\rangle}$.

Example. The standard dot product is an inner product on $\mathbb{F}^n$. (Note also that the real dot product is just the standard dot product on $\mathbb{R}^n$.)

Properties of inner products

Let $\langle\cdot\rangle$ be an inner product on a vector space $(V, \mathbb{F})$.

Lemma. The inner product $\langle\cdot\rangle$ is conjugate linear in the first argument, that is, $\langle \mathbf{v} + \lambda\mathbf{w}, \mathbf{u}\rangle = \langle \mathbf{v}, \mathbf{u}\rangle + \overline{\lambda}\langle \mathbf{w}, \mathbf{u}\rangle$ for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$ and $\lambda \in \mathbb{F}$.

Proof. For all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$ and $\lambda \in \mathbb{F}$, using the fact complex conjugation is additive and multiplicative, we have

$$\langle \mathbf{v} + \lambda\mathbf{w}, \mathbf{u}\rangle = \overline{\langle \mathbf{u}, \mathbf{v} + \lambda\mathbf{w}\rangle} = \overline{\langle \mathbf{u}, \mathbf{v}\rangle + \lambda\langle \mathbf{u}, \mathbf{w}\rangle} = \langle \mathbf{v}, \mathbf{u}\rangle + \overline{\lambda}\langle \mathbf{w}, \mathbf{u}\rangle.$$

Lemma. For all $\mathbf{v} \in V$, we have $\langle \mathbf{0}, \mathbf{v}\rangle = \langle \mathbf{v}, \mathbf{0}\rangle = 0$, and if $\langle \mathbf{u}, \mathbf{v}\rangle = 0$ or $\langle \mathbf{v}, \mathbf{u}\rangle = 0$ for all $\mathbf{u} \in V$ then $\mathbf{v} = \mathbf{0}$.

Proof. Since $\mathbf{0} = 0\mathbf{w}$ for any $\mathbf{w} \in V$, we have $\langle \mathbf{0}, \mathbf{v}\rangle = 0\langle \mathbf{w}, \mathbf{v}\rangle = 0$, and similarly $\langle \mathbf{v}, \mathbf{0}\rangle = \overline{0}\langle \mathbf{v}, \mathbf{w}\rangle = 0$.

Now if $\langle \mathbf{u}, \mathbf{v}\rangle = 0$ or $\langle \mathbf{v}, \mathbf{u}\rangle = 0$ for all $\mathbf{u} \in V$, then in particular $\langle \mathbf{v}, \mathbf{v}\rangle = 0$, in which case $\mathbf{v} = \mathbf{0}$ by positive definiteness.

Lemma. For all $\mathbf{v} \in V$ and $\lambda \in \mathbb{F}$, we have $||\lambda\mathbf{v}|| = |\lambda|\,||\mathbf{v}||$.

Proof. We have $||\lambda\mathbf{v}||^2 = \langle \lambda\mathbf{v}, \lambda\mathbf{v}\rangle = \lambda\overline{\lambda}\langle \mathbf{v}, \mathbf{v}\rangle = |\lambda|^2||\mathbf{v}||^2$. Since all squared terms are non-negative, we can take square roots of both sides, yielding $||\lambda\mathbf{v}|| = |\lambda|\,||\mathbf{v}||$.

Inner product inequalities

Theorem. (Cauchy-Schwarz inequality)
In an inner product space $(V, \mathbb{F})$, we have $|\langle \mathbf{u}, \mathbf{v}\rangle| \leq ||\mathbf{u}||\,||\mathbf{v}||$ for all $\mathbf{u}, \mathbf{v} \in V$.

Proof. Equality is clear when $\mathbf{u}$ or $\mathbf{v}$ is $\mathbf{0}$. Otherwise, for some nonzero $\mathbf{v}, \mathbf{w} \in V$ we have $0 \leq \langle \mathbf{w}-\mathbf{v}, \mathbf{w}-\mathbf{v}\rangle = \langle \mathbf{w}, \mathbf{w}\rangle - \langle \mathbf{w}, \mathbf{v}\rangle - \langle \mathbf{v}, \mathbf{w}\rangle + \langle \mathbf{v}, \mathbf{v}\rangle$ by positive definiteness, so $\langle \mathbf{w}, \mathbf{v}\rangle + \langle \mathbf{v}, \mathbf{w}\rangle \leq ||\mathbf{w}||^2 + ||\mathbf{v}||^2$.

Now setting $\mathbf{w} = \dfrac{\langle \mathbf{u}, \mathbf{v}\rangle}{||\mathbf{u}||^2}\mathbf{u}$ for nonzero $\mathbf{u} \in V$ yields

$$\begin{aligned}
\overline{\tfrac{\langle \mathbf{u}, \mathbf{v}\rangle}{||\mathbf{u}||^2}}\,\langle \mathbf{u}, \mathbf{v}\rangle + \tfrac{\langle \mathbf{u}, \mathbf{v}\rangle}{||\mathbf{u}||^2}\,\overline{\langle \mathbf{u}, \mathbf{v}\rangle} &\leq \left|\tfrac{\langle \mathbf{u}, \mathbf{v}\rangle}{||\mathbf{u}||^2}\right|^2 ||\mathbf{u}||^2 + ||\mathbf{v}||^2, \\
\tfrac{|\langle \mathbf{u}, \mathbf{v}\rangle|^2}{||\mathbf{u}||^2} + \tfrac{|\langle \mathbf{u}, \mathbf{v}\rangle|^2}{||\mathbf{u}||^2} &\leq \tfrac{|\langle \mathbf{u}, \mathbf{v}\rangle|^2}{||\mathbf{u}||^4}\, ||\mathbf{u}||^2 + ||\mathbf{v}||^2, \\
|\langle \mathbf{u}, \mathbf{v}\rangle|^2 &\leq ||\mathbf{u}||^2 ||\mathbf{v}||^2, \\
|\langle \mathbf{u}, \mathbf{v}\rangle| &\leq ||\mathbf{u}||\,||\mathbf{v}|| \quad \text{(all squared terms positive).}
\end{aligned}$$

Lemma. (Triangle inequality)
In an inner product space $(V, \mathbb{F})$, $||\mathbf{u}+\mathbf{v}|| \leq ||\mathbf{u}|| + ||\mathbf{v}||$ for all $\mathbf{u}, \mathbf{v} \in V$.

Proof. We have $||\mathbf{u}+\mathbf{v}||^2 = \langle \mathbf{u}+\mathbf{v}, \mathbf{u}+\mathbf{v}\rangle = ||\mathbf{u}||^2 + \langle \mathbf{u}, \mathbf{v}\rangle + \langle \mathbf{v}, \mathbf{u}\rangle + ||\mathbf{v}||^2$. Note $\langle \mathbf{u}, \mathbf{v}\rangle + \langle \mathbf{v}, \mathbf{u}\rangle = \langle \mathbf{u}, \mathbf{v}\rangle + \overline{\langle \mathbf{u}, \mathbf{v}\rangle} = 2\,\mathrm{Re}(\langle \mathbf{u}, \mathbf{v}\rangle) \leq 2\,|\langle \mathbf{u}, \mathbf{v}\rangle|$. So $||\mathbf{u}+\mathbf{v}||^2 \leq ||\mathbf{u}||^2 + 2|\langle \mathbf{u}, \mathbf{v}\rangle| + ||\mathbf{v}||^2 \leq ||\mathbf{u}||^2 + 2||\mathbf{u}||\,||\mathbf{v}|| + ||\mathbf{v}||^2 = (||\mathbf{u}|| + ||\mathbf{v}||)^2$ by Cauchy-Schwarz, and taking square roots gives the desired result.

Example

Exercise. Show that the function $\langle\cdot\rangle : M_{p,q}(\mathbb{C}) \times M_{p,q}(\mathbb{C}) \to \mathbb{C}$ given by $\langle A, B\rangle = \operatorname{tr}(A^*B)$ for all $A, B \in M_{p,q}(\mathbb{C})$ (where $A^* = \overline{A^T}$) is an inner product.

Proof. Let $A, B, C \in M_{p,q}(\mathbb{C})$ and $\lambda \in \mathbb{C}$. Note that the trace is a linear map on $M_{p,q}(\mathbb{C})$, so

$$\begin{aligned}\langle A, B+\lambda C\rangle = \operatorname{tr}(A^*(B+\lambda C)) = \operatorname{tr}(A^*B + A^*(\lambda C)) &= \operatorname{tr}(A^*B) + \lambda \operatorname{tr}(A^*C) \\ &= \langle A, B\rangle + \lambda\langle A, C\rangle\,,\end{aligned}$$

meaning $\langle\cdot\rangle$ is linear in the second argument.

Next, note that since $\operatorname{tr}(X) = \operatorname{tr}(X^T)$ for any square matrix X, we have

$$\langle A, B\rangle = \operatorname{tr}((A^*B)^T) = \operatorname{tr}(B^T\,\overline{A}) = \operatorname{tr}\left(\overline{(\overline{B^T}A)}\right) = \overline{\operatorname{tr}(B^*A)} = \overline{\langle B, A\rangle},$$

meaning $\langle\cdot\rangle$ is conjugate symmetric.

Finally, note that

$$\langle A, A\rangle = \sum_{i=1}^{q}\sum_{j=1}^{p} A^*_{i,j}A_{j,i} = \sum_{i=1}^{q}\sum_{j=1}^{p} \overline{A_{j,i}}A_{j,i} = \sum_{i=1}^{q}\sum_{j=1}^{p} |A_{j,i}|^2 \geq 0$$

with equality only when $A = \mathbf{0}$, meaning $\langle\cdot\rangle$ is positive definite.

So $\langle\cdot\rangle$ is an inner product.

Example

Exercise. Let $A = \begin{pmatrix} 3 & 2 \\ 2 & 4 \end{pmatrix}$. Show that the function $\langle \cdot \rangle : \mathbb{C}^2 \times \mathbb{C}^2 \to \mathbb{C}$ given by $\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^* A \mathbf{v}$ for all $\mathbf{u}, \mathbf{v} \in \mathbb{C}^2$ is an inner product.

Proof. Let $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathbb{C}^2$ and $\lambda \in \mathbb{C}$. Then

$$\langle \mathbf{u}, \mathbf{v} + \lambda \mathbf{w} \rangle = \mathbf{u}^* A(\mathbf{v} + \lambda \mathbf{w}) = \mathbf{u}^* A \mathbf{v} + \lambda \mathbf{u}^* A \mathbf{w} = \langle \mathbf{u}, \mathbf{v} \rangle + \lambda \langle \mathbf{u}, \mathbf{w} \rangle,$$

meaning $\langle \cdot \rangle$ is linear in the second argument.

Next, noting that $A = A^*$, we have

$$\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^* A \mathbf{v} = (\mathbf{u}^* A \mathbf{v})^T = \mathbf{v}^T A^T \overline{\mathbf{u}} = \overline{\mathbf{v}^* A^* \mathbf{u}} = \overline{\mathbf{v}^* A \mathbf{u}} = \overline{\langle \mathbf{v}, \mathbf{u} \rangle},$$

meaning $\langle \cdot \rangle$ is conjugate symmetric.

Finally, note that writing v_1 and v_2 as the components of $\mathbf{v}$ we have

$$\langle \mathbf{v}, \mathbf{v} \rangle = 3\overline{v_1}v_1 + 2\overline{v_1}v_2 + 2v_1\overline{v_2} + 4\overline{v_2}v_2 = 3|v_1|^2 + 4\,\mathrm{Re}(\overline{v_1}v_2) + 4|v_2|^2.$$

Writing $v_1 = a + bi$ and $v_2 = c + di$ for $a, b, c, d \in \mathbb{R}$, the above becomes

$$3(a^2 + b^2) + 4(ac + bd) + 4(c^2 + d^2) = 2a^2 + 2b^2 + (a + 2c)^2 + (b + 2d)^2 \geq 0$$

with equality only when $\mathbf{v} = \mathbf{0}$, meaning $\langle \cdot \rangle$ is positive definite.

Any matrix A for which $\langle \cdot \rangle$ is an inner product is called positive definite.

More generally, any matrix for which $A = A^*$ is called Hermitian or self-adjoint.

Example

Fact. Let $\mathcal{C}([0,1])$ be the vector space of all continuous functions from the real interval $[0,1]$ to $\mathbb{R}$. The function $\langle \cdot \rangle : \mathcal{C}([0,1]) \times \mathcal{C}([0,1]) \to \mathbb{R}$ given by

$$\langle f, g \rangle = \int_0^1 f(x)g(x)dx$$

for all $f, g \in \mathcal{C}([0,1])$ is an inner product. (Exercise: Prove this!)

Exercise. For all $f, g \in \mathcal{C}([0,1])$, we have

$$\left| \int_0^1 f(x)g(x)dx \right| \leq \sqrt{\int_0^1 f(x)^2 dx \int_0^1 g(x)^2 dx}$$

and

$$\sqrt{\int_0^1 \left(f(x) + g(x)\right)^2 dx} \leq \sqrt{\int_0^1 f(x)^2 dx} + \sqrt{\int_0^1 g(x)^2 dx}\,.$$

Proof. These are direct applications of the Cauchy-Schwarz and triangle inequalities to the inner product defined above.

Note these would be much harder to prove without the inner product connection!